EIS analysis report

Model: Randles (Rs + Rct‖Cdl)
Data file: randles_Rs10_Rct100_C10uF.csv
Generated on 2026-10-10 19:09

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1. Equivalent-circuit fit

Electrical parameters of the chosen model, found by least-squares fitting. They describe the circuit, not yet the chemistry.

R_s: 10.01 Ω
R_ct: 99.94 Ω
C_dl: 9.994e-06 F

RMSE (complex): 0.1801 Ω
χ² (σ = 1): 9.606

2. Measured EIS features (no model)

Read directly from the data. If the system behaves like a Randles circuit: R_high ≈ R_s, R_low − R_high ≈ R_ct, and f_apex relates to R_ct and C_dl as below.

R_high: 10.19 Ω
R_low: 110.1 Ω
R_low − R_high: 99.96 Ω
f_apex: 154.5 Hz

```
f_{apex} = \frac{1}{2 \pi \, R_{ct} \, C_{dl}}
```

3. Electrochemical interpretation

Derived quantities. Each one is valid only under the assumptions listed with it. Names, units and formulas are written in LaTeX (between $...$) so they can be pasted into a document.

Inputs provided

- Temperature: 25 °C
- Electrode area: 1 cm²
- Number of electrons n: 1

Time constant τ

Value: 0.0009988 s

LaTeX formula:

```
\tau = R_{ct} \, C_{dl}
```

Assumptions:

- The fitted circuit is $R_s + (R_{ct} / C_{dl})$: one semicircle, and C_{dl} is an ideal capacitor (no CPE / non-ideal behaviour).

Notes:

- Describes how fast the interface charges/discharges. It is a property of the fit, not yet a chemical statement.

Characteristic frequency f_c

Value: 159.3 Hz

LaTeX formula:

```
f_c = \frac{1}{2 \pi \tau}
```

Assumptions:

- The fitted circuit is $R_s + (R_{ct} / C_{dl})$: one semicircle, and C_{dl} is an ideal capacitor (no CPE / non-ideal behaviour).

Notes:

- Should match the apex frequency measured directly from the Nyquist plot (f_{apex}); a large mismatch means the single-semicircle model does not describe the data well.

Specific double-layer capacitance

Value: 9.994e-06 F/cm^2

LaTeX formula:

```
C_{spec} = \frac{C_{dl}}{A}
```

Assumptions:

- The fitted circuit is $R_s + (R_{ct} / C_{dl})$: one semicircle, and C_{dl} is an ideal capacitor (no CPE / non-ideal behaviour).
- A is the true electroactive area, not just the geometric area.
- The fitted capacitance is really the double layer (not adsorption pseudocapacitance or a coating).

Exchange current density j_0

Value: 0.0002571 A/cm^2

LaTeX formula:

```
j_0 = \frac{R\,T}{n\,F\,R_{ct}\,A}
```

Assumptions:

- The fitted circuit is $R_s + (R_{ct} / C_{dl})$: one semicircle, and C_{dl} is an ideal capacitor (no CPE / non-ideal behaviour).
- Measured at equilibrium (open-circuit) with both the oxidised and reduced species present, small AC amplitude (linearised Butler-Volmer).
- R_{ct} is purely the charge-transfer resistance: no diffusion (Warburg) or film resistance mixed into it.
- Single, simple electron-transfer reaction; n is the number of electrons of that reaction (ambiguous for multi-step mechanisms).
- A is the true electroactive area.
- $T = {T_kelvin:.2f}$ K.

Notes:

- If R_{ct} also contains other resistive contributions, j_0 is underestimated.

Consistency check: fit vs data

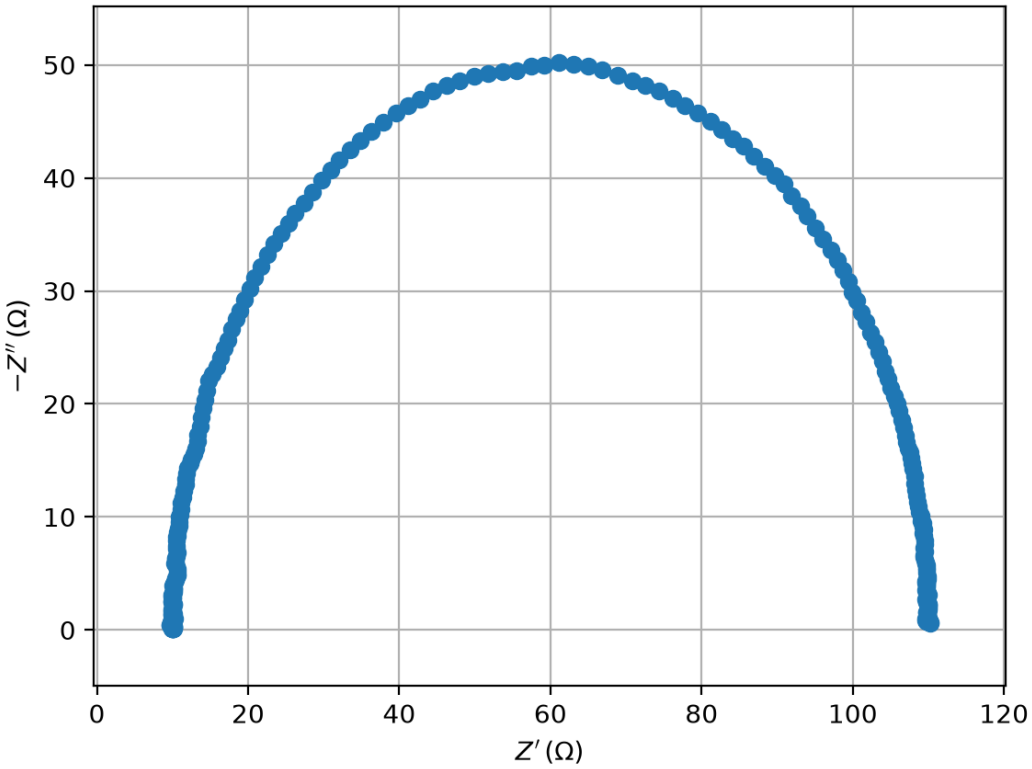
f_c (fit): 159.3 Hz

f_apex (data): 154.5 Hz

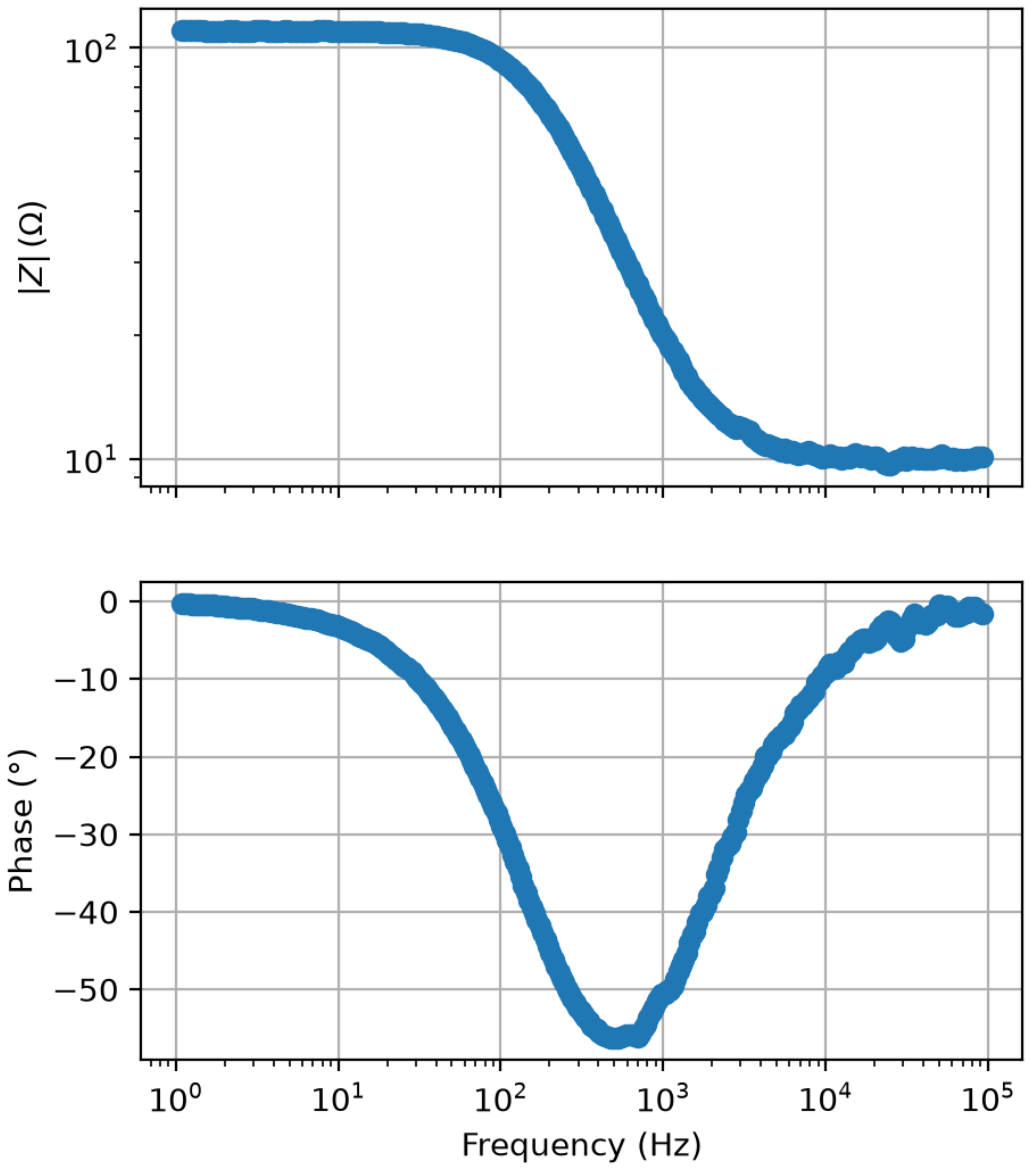
Difference: 3.1 %

The fit and the data agree on the time constant. This supports the single-semicircle model but does not prove it.

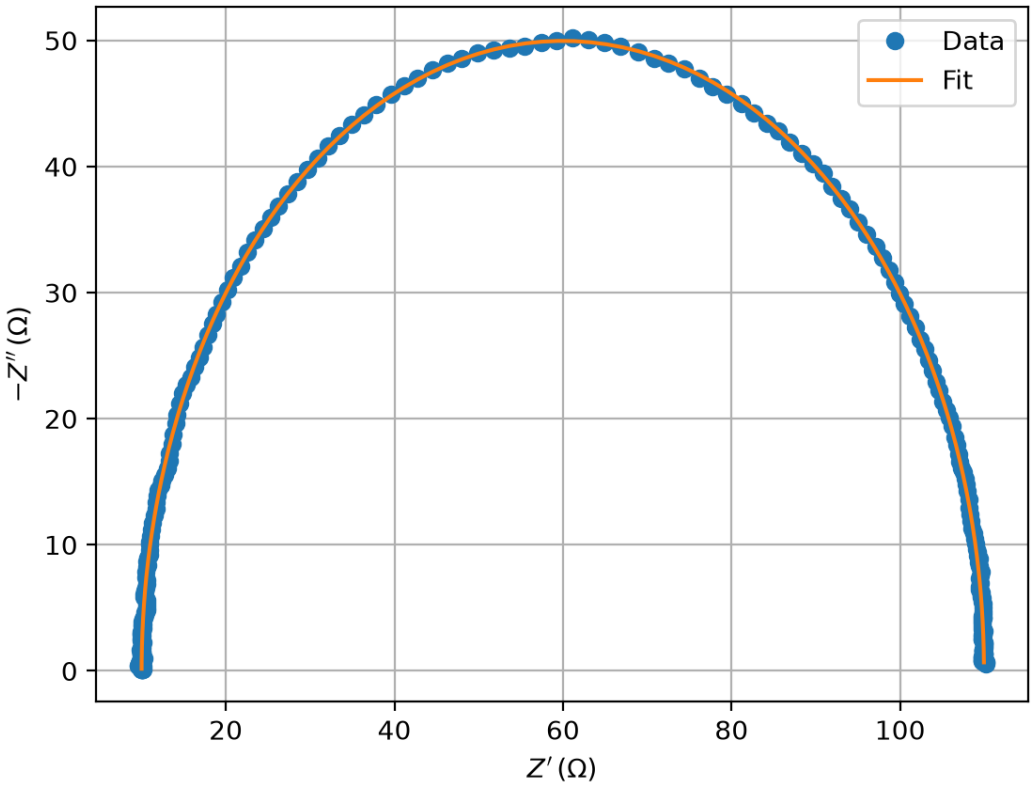
4. Figures



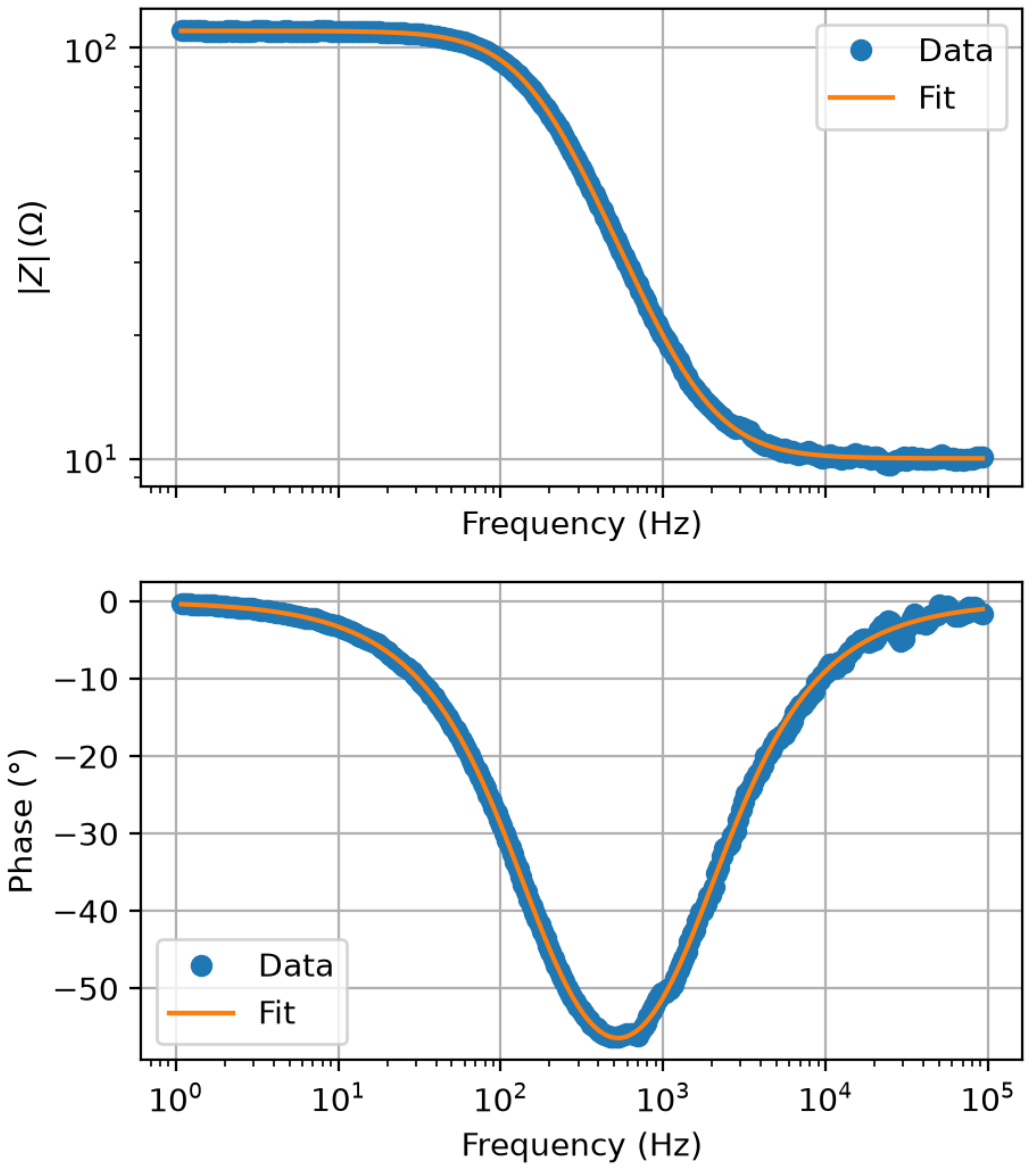
Nyquist plot (data)



Bode plot (data)



Nyquist: data vs fitted model



Bode: data vs fitted model